Fabio Furini

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Part I – General Information

Date of Birth	**12 January 1982**
Nationality	**Italian**
Homepage	`https://fabiofurini.github.io`
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Last updated	**November 7, 2026**

At a glance

62	**31**	**22**
Refereed publications	**Google Scholar h-index**	**Scopus h-index**
50 journal papers; 12 proceedings		
7	**∼ €800k**	**Several**
Researchers supervised	**Research funding**	**Industrial collaborations**
6 PhD students; 1 postdoctoral researcher	grants and contracts	Production planning; telecommunications; network robustness; rail transport

Part II – Education

2017 **HDR: Habilitation à diriger des recherches**, highest academic degree in the French university system. Title: "Reformulations and Decompositions of Mixed Integer Linear and Nonlinear Programs", Université Paris Dauphine-PSL, France. Jury: Professor Mauro Dell'Amico, Professor Martine Labbé, Professor Nelson Maculan, Professor Ridha Mahjoub, Professor Vangelis Pascos, Professor Eduardo Uchoa and Professor Roberto Wolfler Calvo.

2008–2011 **Ph.D. in Control System Engineering and Operational Research**, title: "Decomposition and reformulation of integer linear programming problems". University of Bologna, Italy. Supervisors: Professor Paolo Toth and Professor Alberto Caprara.

2005–2007 **Master's degree in Engineering and Industrial Management**, University of Bologna, Italy.

2001–2004 **Bachelor's degree in Engineering and Industrial Management**, University of Bologna, Italy.

Academic qualifications

2022 **Qualification as Full Professor in Operations Research**, from the Assessment Committee of the Technical University of Denmark in the Danish academic system.

2019 **Qualification as Full Professor in Operations Research**, (MAT/09 - 01/A6) from the Italian Ministry of University and Research (MIUR) ("art. 16, legge 240/2010"), BANDO D.D. n. 1532/2016.

2017 **French national qualification as Professeur des universités**, national qualification to apply for Full Professor positions in the French academic system, distinct from the HDR degree. Qualification N° 20126246615 in section 26-Mathématiques appliquées et applications des mathématiques; N° qualification 18127246615 in section 27-Informatique.

Spoken languages

Italian, French, English.

Part III – Appointments

2021–now **Associate Professor**, Department of Computer, Control and Management Engineering, Sapienza University of Rome.

Academic and other appointments

2021–2026 **Associate Researcher**, Italian National Research Council (IASI-CNR), Rome.

2020–2021 **Research fellow**, Italian National Research Council (IASI-CNR), Rome.

2013–2019 **Maître de conférences**, (equivalent to Associate Professor) Laboratoire d'Analyse et Modélisation de Systèmes pour l'Aide à la Décision (LAMSADE), Université Paris Dauphine-PSL, UMR-CNRS 7243.

2012–2013 **Postdoctoral Fellow**, Laboratoire d'Informatique de Paris-Nord (LIPN), Sorbonne Paris Nord University (formerly Université Paris 13), UMR-CNRS 7030.

2011–2012 **Postdoctoral Fellow**, *University of Bologna, Italy.*

2009–2012 **Consultant, Ferrovie dello Stato**, *Rome, Italy.*

2007–2008 **Consultant, Ferrari S.p.A.**, *Maranello (Modena), Italy.*

Part IV – Teaching Experience

Academic Year	University	Hours
2025-2026	Sapienza University of Rome	120
2024-2025	Sapienza University of Rome	120
2023-2024	Sapienza University of Rome / University of Modena and Reggio Emilia	230
2022-2023	Sapienza University of Rome	150
2021-2022	Sapienza University of Rome	120
2019-2020	Université Paris Dauphine-PSL	132
2018-2019	University of Parma / Université Paris Dauphine-PSL	192
2017–2018	Université Paris Dauphine-PSL	138
2016–2017	Université Paris Dauphine-PSL	102
2015-2016	Université Paris Dauphine-PSL	128
2014-2015	Université Paris Dauphine-PSL	83
2013-2014	Université Paris Dauphine-PSL	93
2012-2013	Université Paris Dauphine-PSL / University of Bologna	56
Total		**1644**

2025-2026

Programmazione intera e ottimizzazione combinatoria, Sapienza University of Rome, *master's degree, 120 hours, 300 students.*

2024-2025

Programmazione intera e ottimizzazione combinatoria, Sapienza University of Rome, *master's degree, 120 hours, 300 students.*

2023-2024

Programmazione intera e ottimizzazione combinatoria, Sapienza University of Rome, *master's degree, 90 hours, 300 students.*

Analisi matematica 1, Sapienza University of Rome, *bachelor's degree, 90 hours, 150 students.*

Ottimizzazione lineare intera, University of Modena and Reggio Emilia, *bachelor's degree, 50 hours, 100 students.*

2022-2023

Combinatorial Optimization and Integer Programming, Sapienza University of Rome, *master's degree, 60 hours, 100 students.*

Analisi matematica 1, Sapienza University of Rome, *bachelor's degree, 90 hours, 150 students.*

2021-2022

Combinatorial Optimization and Integer Programming, Sapienza University of Rome, *master's degree, 60 hours, 100 students.*

Analisi matematica 1, Sapienza University of Rome, *bachelor's degree, 60 hours, 150 students.*

2019-2020

Outils d'optimisation pour les sciences des données et de la décision, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Gestion de la production, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Système de gestion de BDD, Université Paris Dauphine-PSL, *master's degree, 42 hours, 40 students.*

Optimisation en finance, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Programmation linéaire, Université Paris Dauphine-PSL, *bachelor's degree, 24 hours, 40 students.*

2018-2019

Quantitative Methods and Optimization, University of Parma, *master's degree, 60 hours, 10 students.*

Outils d'optimisation pour les sciences des données et de la décision, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Gestion de la production, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Système de gestion de BDD, Université Paris Dauphine-PSL, *master's degree, 42 hours, 40 students.*

Optimisation en finance, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Programmation linéaire, Université Paris Dauphine-PSL, *bachelor's degree, 24 hours, 40 students.*

2017–2018

Outils d'optimisation pour les sciences des données et de la décision, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Gestion de la production, Université Paris Dauphine-PSL, *master's degree, 54 hours, 40 students.*

Optimisation en finance, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Programmation linéaire avancée, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Optimisation combinatoire, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

2016–2017

Outils d'optimisation pour les sciences des données et de la décision, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Gestion de la production, Université Paris Dauphine-PSL, *master's degree, 54 hours, 40 students.*

Optimisation en finance, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

2015-2016

Outils d'optimisation pour les sciences des données et de la décision, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Gestion de la production, Université Paris Dauphine-PSL, *master's degree, 36 hours, 40 students.*

Optimisation en finance, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Humanitarian logistics, MSO, Université Paris Dauphine-PSL, *master's degree, 9 hours, 40 students.*

Programmation linéaire, Université Paris Dauphine-PSL, *bachelor's degree, 17 hours, 40 students.*

Système de gestion de BDD, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

2014-2015

Optimisation en finance, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Humanitarian logistics, MSO, Université Paris Dauphine-PSL, *master's degree, 9 hours, 40 students.*

Programmation linéaire, Université Paris Dauphine-PSL, *bachelor's degree, 17 hours, 40 students.*

Système de gestion de BDD, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Résolution de problèmes de grande taille et mise en œuvre informatique, Université Paris Dauphine-PSL, *master's degree, 15 hours, 40 students.*

2013-2014

Applications de l'optimisation combinatoire, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Algo et programmazione JAVA, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Optimisation en finance, Université Paris Dauphine-PSL, *master's degree, 24 hours, 40 students.*

Résolution de problèmes de grande taille et mise en œuvre informatique, Université Paris Dauphine-PSL, *master's degree, 15 hours, 40 students.*

Système de gestion de BDD, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

2012-2013

Système de gestion de BDD, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Algo et programmazione JAVA, Université Paris Dauphine-PSL, *master's degree, 18 hours, 40 students.*

Optimization software and MIP solvers, University of Bologna, *bachelor's degree, 20 hours, 40 students.*

Teaching in PhD courses

2023 **CIMPA Research School on Graph Structure and Complex Network Analysis**, Integer Programming Formulations and Exact Algorithms for Hard Combinatorial Optimization Problems on Graphs, *Nesin Mathematics Village, Şirince, İzmir, Turkey.*

2019 **Benders Decomposition**, University of Bologna, Italy.

2019 **Dantzig-Wolfe Decomposition**, IFSTAR, Lille.

2018 **Modeling and Solving Decision Problems using Integer Programs with applications in Finance, Data Science and Production Management**, University of Parma, Italy.

2015 **Column generation for hard combinatorial problems**, Université Paris Dauphine-PSL.

Teaching material and online course resources

All course material (lecture notes, models, code, data and exercises) is openly accessible online, in English and Italian, and collected at `https://fabiofurini.github.io/teaching/`. Source repositories: `https://github.com/fabiofurini`.

2026 **Integer Linear Programming / Programmazione lineare intera**, lecture notes with worked examples and exercises on integer linear programming models, polyhedral methods and exact algorithms, *Course sites:* `integer-linear-programming` *(EN)* · `programmazione-lineare-intera` *(IT).*

2026 **Combinatorial Optimization / Ottimizzazione combinatoria**, lecture notes with worked examples and exercises on combinatorial optimization problems and algorithms, *Course sites:* `combinatorial-optimization` *(EN)* · `ottimizzazione-combinatoria` *(IT).*

2026 **MIP Modelling / Modellazione MIP**, mixed-integer linear programming models for Management Engineering, with modelling exercises, *Course sites:* `mip-modelling` *(EN)* · `modellazione-mip` *(IT).*

2026 **Operations Research Lab / Laboratorio di ricerca operativa**, continuous optimization models (LP, QP and NLP) and case studies, with lecture notes, Python/Gurobi code, data and slides, *Course sites:* `operations-research-lab` *(EN)* · `laboratorio-ricerca-operativa` *(IT).*

2026 **Network Optimization / Ottimizzazione su reti**, graph algorithms for spanning trees, shortest paths, flows, transportation, assignment and project planning, with worked examples and exercises, *Course sites:* `network-optimization` *(EN)* · `ottimizzazione-su-reti` *(IT).*

2026 **Linear Programming / Programmazione lineare**, linear programming, the simplex method and duality, with exercises and worked solutions, *Course sites:* `linear-programming` *(EN)* · `programmazione-lineare` *(IT).*

2026 **Convex Optimization / Ottimizzazione convessa**, convex sets and functions, optimality conditions and algorithms, with exercises and interactive graphs, *Course sites:* `convex-optimization` *(EN)* · `ottimizzazione-convessa` *(IT).*

2026 **Algorithms / Algoritmi**, algorithm design and analysis: sorting, searching, asymptotic notation and recursion, with worked examples, exercises and an interactive lab, *Course sites:* `algorithms` *(EN)* · `algoritmi` *(IT).*

2026 **Linear Algebra / Algebra lineare**, vectors, matrices, linear systems and eigenvalues, with online lecture notes and an interactive computation lab, *Course sites:* `linear-algebra` *(EN)* · `algebra-lineare` *(IT).*

2026 **Mathematical Analysis 1 / Analisi matematica 1**, online lecture notes for bachelor-level mathematics: numbers and logic, functions, limits, derivatives and series, with proofs, examples and exercise sets, *Course sites:* `mathematical-analysis-1` *(EN)* · `analisi-matematica-1` *(IT).*

Bomberino, an interactive workbench for teaching operations research algorithms (with Fabio Ciccarelli): primal and dual simplex, Gomory cutting planes and branch-and-bound on small linear and integer programs, solved in exact fractions with every pivot, bound, cut and branching step recorded, and exportable to LaTeX, `fciccarelli.com/bomberino`

Part V – Society Memberships, Awards and Honors

Current **Member**, Mathematical Optimization Society.

2014–2021 **Prime d'encadrement doctoral et de recherche (PEDR)**, French academic distinction awarded following a positive evaluation of scientific production, doctoral and scientific supervision, dissemination, and scientific responsibilities.

2014 **Prix Fondation Dauphine de la Publication**, Université Paris Dauphine-PSL.

Part VI – Funding Information

Technology-transfer context: The industrial doctoral contracts listed below are French industrial PhD projects designed to transfer academic research into company-driven innovation. In these collaborations, the research questions came from industrial needs and the outputs included mathematical models, algorithms, computational benchmarks, software prototypes and decision-support tools for operational use.

2025 **"Towards EXplainable BIlevel Learning"**, Sapienza University of Rome, Progetti di Ricerca Medi, *role: co-principal investigator. Protocol: RM125199C3C9A6C5.*

2024 **"Proactive Anomaly Network Defense and optimization Analysis (PANDA)"**, Sapienza University of Rome, Progetti di Ricerca (Piccoli, Medi) – Progetti Medi, *role: principal investigator. Protocol: RM124190A6B4432F. Project team: 1 other participant, 2 faculty/research staff, 2 PhD students/research fellows/specialists.*

2023 **"SEEDS: Sustainable Ecosystems in Evolving Digital Societies"**, Sapienza University of Rome, Progetti Dipartimentali, *role: project member. Protocol: RD12318A97C04F22. Aggregate project data: 29 faculty/research staff involved; funded amount about EUR 87k. My share was about one quarter of the funding and supported a one-year research fellowship.*

2023–now **"Rail Transportation - CN4 - Spoke 4"**, Centro Nazionale 4, PNRR – Spoke 4, *role: co-principal investigator. Protocol: CN422184599DB73C. Person-month commitment: 9. Aggregate project data: 4 thematic lines, 12 research participants, proposed funding about EUR 1623k. My DIAG share was about EUR 151k, including about EUR 126k for critical-mass personnel, within the thematic line "Increase of capacity of railway transport".*

2021 **"Optimal spread-containment policies to protect social networks"**, SEED PNR, *role: principal investigator.*

2017–2018 **"Rich Graph Coloring Problems with applications to resource allocation"**, Projet PGMO, *role: principal investigator.*

2016–2017 **"Efficient Quadratic Programming Exact Algorithms mixing Relaxations"**, PEPS JCJC of the CNRS, *role: principal investigator.*

2014–2017 **"QPLIB2014: a Library of Quadratic Programming Instances"**, Projet PGMO, *role: principal investigator.*

2014–2017 **"Efficient exact algorithms for Graph Partitioning Problems"**, Projet PGMO, *role: principal investigator.*

2015–2017 **Industrial doctoral contract**, DecisionBrain, Paris, *technology-transfer focus: optimization-based production planning and scheduling for apparel manufacturing. The collaboration translated mixed-integer programming, decomposition and heuristic methods into a software-oriented decision-support system for industrial lot-sizing, parallel-machine planning, setup carryover and production sequencing.*

2017–2019 **Industrial doctoral contract**, Orange, Paris, *technology-transfer focus: optimization methods for next-generation telecommunication networks. The work addressed Virtual Network Functions placement and routing in Software Defined Networks, producing MILP formulations, valid inequalities, heuristics and branch-and-price algorithms for latency-constrained service-chain design.*

2020–2022 **Industrial doctoral contract**, Huawei, Paris, *technology-transfer focus: network optimization, interdiction and robustness analysis. The project developed polyhedral models and algorithms for maximum-flow and multicommodity-flow blocker problems, supporting industrial evaluation of vulnerable components, critical flows and protection strategies in large networked systems.*

2010–2012 **Project Rail on Timetabling Optimization Software**, with Ferrovie dello Stato (Italian national railway company).

	year												total in k€
Project	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	
DecisionBrain industrial PhD	45	45	45										135
Orange industrial PhD			45	45	45								135
Huawei industrial PhD						55	55	55					165
PEPS-CNR		15											15
SEED PNR								10			10		20
PGMO	10	10	10	10									40
Invited Professors	5	5	5	5	10	5						5	35
PostDoc											40		40
PNRR									51	50	50		151
SEEDS									22				22
PANDA										10			10
Bilevel Learning											19		19
approx. total in k€	60	75	105	60	55	60	55	65	73	60	119	5	**∼800**

Part VII – Research Activities

Mixed Integer Linear Programming (MILP), study and development of algorithms for mixed integer linear programming problems. My research focuses on advanced techniques including polyhedral investigations, branch-and-cut algorithms for cutting plane generation, and Benders decomposition for problem-specific reformulations. These methodologies enable the exact solution of complex real-world optimization problems.

Column Generation and Decomposition Methods, study and development of advanced decomposition-based algorithms founded on column generation for Dantzig–Wolfe reformulations and branch-and-price approaches. Research includes a focus on automated decomposition detection, reformulation strategies to obtain strong dual bounds, and implementation of these techniques within state-of-the-art optimization solvers.

Graph Optimization Problems, development of exact and heuristic algorithms for a wide range of graph-based optimization problems, including vertex coloring, partition and covering problems, network design and graph partitioning. These problems are ubiquitous in modern applications spanning scheduling, routing and resource allocation.

Cutting and Packing Problems, design and analysis of algorithms for cutting stock problems, bin packing and knapsack problems in their various forms. The research combines mathematical programming formulations with advanced decomposition techniques and dynamic programming approaches to achieve optimal or near-optimal solutions with small optimality gaps.

Dynamic Programming and Combinatorial Optimization, study and development of dynamic programming algorithms for hard combinatorial optimization problems. The focus is on designing efficient state-space representations and transition mechanisms to solve large-scale instances arising from practical applications in scheduling, routing and resource planning.

Real-world Applications, development and application of optimization methodologies to important practical problems including train timetabling in railway networks, air traffic management, production planning and scheduling, vehicle routing and logistics, facility location and network design problems. These applications drive the development of new algorithms and problem formulations.

Technology transfer and industrial collaborations

Optimization software, my technology-transfer activity is documented by industrial doctoral projects, research contracts and software-oriented collaborations with companies and public industrial stakeholders. The common outcome is the translation of mathematical optimization research into operational models, algorithms, computational tools and reproducible software components for planning, scheduling, network design and resource allocation.

DecisionBrain	Industrial doctoral collaboration on production planning and scheduling in apparel manufacturing. The project connected real production constraints with mixed-integer programming, decomposition algorithms and heuristics, leading to a software-oriented planning system and computational benchmarks for industrial lot-sizing and sequencing decisions.
Orange	Industrial doctoral collaboration on Virtual Network Functions placement and routing for Software Defined Networks. The work supported technology transfer toward telecommunications network planning through optimization models and algorithms for latency-constrained service-chain deployment.
Huawei	Industrial doctoral collaboration on flow-blocker and interdiction problems for large networked systems. The work connected bilevel and polyhedral optimization with industrial questions on network robustness, critical-flow identification and protection strategies.
Rail transport	Industrial and PNRR-funded activity on railway optimization, including train timetabling and capacity-increase problems. The research transfers exact and heuristic optimization methods into decision-support tools for rail operations and infrastructure-capacity planning.

Research Methodology, my research integrates mathematics, economics and information technology to develop methodologies for solving complex optimization problems in practice. The approach combines the design of effective mathematical models with the development of sophisticated exact and heuristic algorithms. Key methodologies include column generation, branch-and-price, branch-and-cut, Benders decomposition, cutting planes, dynamic programming and polyhedral theory.

Algorithm development and dissemination

Open-source algorithm implementations, development and dissemination of exact algorithms, benchmark instances and computational material for combinatorial optimization, *the public material covers implementations for maximum clique problems, binary constraint satisfaction, knapsack problems with conflicts, edge-weighted maximum clique, location covering problems, submodular utility maximization, the k-vertex cut problem, two-dimensional knapsack problems and Ramsey-number lower bounds. Web page:* `https://fabiofurini.github.io/software/`

CliSAT	Exact SAT-based algorithm for the Maximum Clique Problem. It is used to solve hard maximum clique instances and to disseminate benchmark instances and computational material.

P. San Segundo, F. Furini, D. Alvarez, P.M. Pardalos. *CliSAT: A New Exact Algorithm for Hard Maximum Clique Problems.* · github.com/psanse/CliSAT

Bfilt Branch-and-filter exact algorithm for Binary Constraint Satisfaction Problems. It provides a Linux release, benchmark instances and format converters for reproducing computational experiments on binary CSP instances.

P. San Segundo, F. Furini, R. León. *A New Branch-and-Filter Exact Algorithm for Binary Constraint Satisfaction Problems.* · github.com/psanse/Bfilt

KPCG CFS combinatorial branch-and-bound algorithm for the Knapsack Problem with Conflicts. It solves knapsack instances in which selected items must satisfy both a capacity constraint and pairwise incompatibility constraints.

S. Coniglio, F. Furini, P. San Segundo. *A New Combinatorial Branch-and-Bound Algorithm for the Knapsack Problem with Conflicts.* · github.com/psanse/KPCG

BBEWC Exact branch-and-bound algorithm for the Maximum Edge Weighted Clique Problem. It is used to find a clique maximizing the total weight of its edges and includes the datasets and Linux executable used in the computational study.

P. San Segundo, S. Coniglio, F. Furini, I. Ljubic. *A New Branch-and-Bound Algorithm for the Maximum Edge-Weighted Clique Problem.* · github.com/psanse/BBEWC

LOC-COV Branch-and-Benders-cut software for the Maximal Covering Location Problem and the Partial Set Covering Location Problem. It supports large-scale facility-location and covering applications where demand must be covered under distance, budget or coverage constraints.

J.-F. Cordeau, F. Furini, I. Ljubic. *Benders Decomposition for Very Large Scale Partial Set Covering and Maximal Covering Location Problems.* · github.com/fabiofurini/LocationCovering

SUB-COV-MAX Optimization code for submodular maximization of concave utility functions composed with a set-union operator. It is designed for covering and location models with diminishing-return utility functions and relies on CPLEX callable libraries.

S. Coniglio, F. Furini, I. Ljubic. *Submodular Maximization of Concave Utility Functions Composed with a Set-Union Operator with Applications to Maximal Covering Location Problems.* · github.com/fabiofurini/Maximizing-submodular-utility-functions

BP-K-VCP Branch-and-price software for the k-vertex cut problem. It is used to find a minimum-cost set of vertices whose removal disconnects a graph into at least k connected components, with applications in network interdiction and critical-node detection.

F. Ciccarelli, F. Furini, C. Hojny, M.E. Lübbecke. *Branch-and-Price Strikes Back for the k-Vertex Cut Problem.* · github.com/FabioCiccarelli/BP_k-vertex-cut

NEA-G2KP Numerically exact and non-numerically exact algorithms for two-dimensional knapsack and packing problems. The repository contains source code, instances, results and solution data for 2KP, 2KPC, 2OPP and G2KP benchmark studies.

S. Wang, R. Baldacci, F. Furini, L. Wei, Q. Liu. *Exact Algorithms for Two-Dimensional Knapsack Problems: A Unified Framework with New Benchmark Results.* · github.com/INFORMSJoC/2025.1423

BPP Numerically Exact Software and computational material for a numerically exact algorithm for the Bin-Packing Problem.

R. Baldacci, S. Coniglio, F. Furini. *A Numerically Exact Algorithm for the Bin-Packing Problem.* · github.com/fabiofurini/bpp-numerically-exact

Ramsey Number Lower Bounds Computational package for searching new lower bounds on Ramsey numbers and computing circulant Ramsey numbers. It includes executable optimization models, an independent (m, n)-coloring certificate checker, and graph certificates supporting the reported lower bounds.

S. Coniglio, F. Furini, I. Ljubić, P. San Segundo, J. Thürauf, E. Traversi. *An Integer Programming Approach to Compute Lower Bounds for Ramsey Numbers Using Circulant Graphs.* · github.com/fabiofurini/ramsey-number-lower-bounds

Classification of the publications

The two tables below map my research onto the full list of publications in Part IX; labels in square brackets follow that numbering, and a paper may appear in more than one category. The first table groups the 50 refereed journal papers by research topic, following the same topics used on my personal website, and the second groups them by the main methodological technique employed.

Publications by research topic

Research topic	Publications	Note
Bin Packing & Cutting Stock	[2], [4], [12], [31], [32], [40], [42], [44], [50]	Branch-price-and-cut, pattern enumeration and numerically safe dual bounds, including setups and the temporal and two-dimensional variants.
Knapsack Problems	[3], [8], [10], [16], [18], [20], [34], [38], [46]	Knapsack variants with setups, conflicts, products and time windows, together with the polyhedral study of cover inequalities.
Graph Colouring	[1], [14], [15], [21], [30], [35]	Branch-and-price and DSATUR-based branch-and-bound for vertex, weighted, partition and minimum-sum colouring, including reformulations as maximum weight stable set problems.
Clique & Stable Set	[27], [28], [33], [39], [43], [48]	Exact combinatorial algorithms for maximum clique and its weighted and edge-weighted variants, with SAT-based filtering, tightened upper bounds and relaxed-clique decompositions.
Interdiction, Blocker & Vertex Cut	[22], [26], [29], [36], [37], [45], [49]	Bilevel and interdiction problems on graphs: vertex k-cut, capacitated vertex separators, clique and edge interdiction, and maximum flow blockers.
Transportation & Scheduling	[7], [9], [13], [19], [47]	Real-world optimisation in air and rail traffic: aircraft sequencing, train timetabling and routing, and barge container ship routing.
Decomposition & Reformulation	[6], [11], [17]	Automatic Dantzig–Wolfe reformulation and perspective relaxations for mixed integer programs with semi-continuous variables.
Covering, Location & Submodularity	[25], [41]	Benders decomposition for large-scale covering and location problems, plus submodular maximization applications.
Binary Quadratic Programming	[23], [24]	Linearisation techniques for binary quadratic problems and QPLIB, the reference library of quadratic programming instances.
Other	[5]	Generation of antipodal random vectors with prescribed second-order statistics (signal processing).

Publications by methodological technique

Technique	Publications	Note
Column generation, Dantzig–Wolfe reformulation and branch-and-price	[1], [2], [3], [6], [10], [31], [32], [33], [35], [44], [49]	Dantzig–Wolfe reformulations and set-partitioning models solved by pricing; automatic detection of decomposable structure; recursive and lexicographic pricing schemes.
Combinatorial branch-and-bound	[15], [18], [27], [28], [34], [39], [43], [48]	Problem-specific bounding and branching rules, combinatorial bounds and filtering, benchmarked against general-purpose solvers.
Branch-and-cut, cutting planes and polyhedral study	[21], [22], [37], [38], [45]	Valid inequalities and exact separation procedures; polyhedral analysis of integer formulations and their strengthening.
Benders decomposition and bilevel optimization	[25], [26], [29], [36], [37], [45]	Classical and combinatorial Benders decomposition for large-scale models; bilevel reformulations of interdiction, blocker and separator problems.

Dynamic programming	[8], [16], [18], [42]	State-space representations and transition mechanisms for knapsack and packing problems, also used as bounding and pricing procedures.
Integer programming formulations and modelling	[4], [5], [12], [13], [14], [19], [20], [46], [47], [50]	Design and comparison of alternative MILP formulations, including tighter models and reformulations of a problem as a different combinatorial problem.
Reformulations and relaxations for quadratic and nonlinear programs	[11], [17], [23], [24], [41]	Perspective and projected perspective reformulations, linearisation techniques for binary quadratic problems, and submodular objectives.
Heuristics, matheuristics and problem reduction	[2], [7], [8], [9], [40]	Rolling-horizon and column-generation heuristics for real-world instances; variable and constraint reduction techniques.
Structural and complexity analysis	[20], [30], [38], [50]	Complexity results and structural properties supporting the design of exact algorithms.
Numerically exact algorithms, libraries and software	[6], [23], [44]	Numerically safe exact algorithms, open instance libraries and solver-oriented implementations supporting reproducible computational work.

Part VIII – Summary of Scientific Achievements

62	**2,499**	**31**	**∼ €800k**
Refereed publications	Google Scholar citations	Google Scholar h-index	Research funding

Publication record

Indicator	Value	Detail
Refereed publications	**62**	50 international journal papers; 12 conference proceedings
Working papers and preprints	**5**	current manuscripts and submitted papers

Bibliometric profile

Database	Citations	h-index	i10-index	Additional data
Google Scholar	**2,499**	**31**	**48**	all years
Google Scholar	**1,703**	**24**	**42**	since 2021
Scopus	**1,429**	**22**	–	61 documents; 1,159 citing documents

Research impact and collaboration

Indicator	Value	Detail
Highly cited documents	**20**	documents in the top 25% most cited worldwide
Top CiteScore journal documents	**25**	documents in journals in the top 25% by CiteScore percentile
Field-Weighted Citation Impact	**0.97**	SciVal author impact indicator
International collaboration	**38**	internationally co-authored documents
Co-author network	**71**	Scopus co-authors; 12 countries; 25 institutions
SciVal research topics	**9**	topics indexed for the author profile

Funding, software and technology transfer

Indicator	Value	Detail
Research funding through grants and contracts	∼ **€800k**	includes grants, contracts and invited-professor funding
Rail-transport research funding share	**€151k**	capacity increase and rail-transport optimization
SEEDS research funding share	**€22k**	supported one one-year research fellowship
Public software repositories	**10**	repositories connected to publications and reproducible computational work
Technology-transfer projects	**4**	DecisionBrain, Orange, Huawei and rail transport
Industrial doctoral projects with companies	**3**	industrial doctoral collaborations with DecisionBrain, Orange and Huawei
Supervision	**7**	6 PhD students supervised or co-supervised; 1 postdoctoral researcher supervised

Most cited publications

Ref.	Year	Title	Journal	Cit.
[25]	2019	Benders Decomposition for Very Large Scale Partial Set Covering and Maximal Covering Location Problems	Eur. J. Oper. Res.	**128**
[9]	2016	Approaches to a real-world train timetabling problem in a railway node	Omega	**103**
[23]	2019	QPLIB: A Library of Quadratic Programming Instances	Math. Program. Comput.	**90**
[7]	2015	Improved rolling horizon approaches to the aircraft sequencing problem	J. Sched.	**88**
[34]	2021	A new combinatorial branch-and-bound algorithm for the Knapsack Problem with Conflicts	Eur. J. Oper. Res.	**84**
[32]	2020	A branch-and-price algorithm for the temporal bin packing problem	Comput. Oper. Res.	**64**
[2]	2012	A Column Generation Heuristic for the Two-Dimensional Two-Staged Guillotine Cutting Stock Problem with Multiple Stock Size	Eur. J. Oper. Res.	**62**
[12]	2016	Modeling Two-Dimensional Guillotine Cutting Problems via Integer Programming	INFORMS J. Comput.	**60**
[6]	2015	Automatic Dantzig-Wolfe Reformulation of Mixed Integer Programs	Math. Program.	**51**
[4]	2013	Models for the Two-Dimensional Two-Stage Cutting Stock Problem with Multiple Stock Size	Comput. Oper. Res.	**49**

References in square brackets follow the numbering of the full list of publications in Part IX. Citation counts from OpenAlex (`https://openalex.org`), ORCID 0000-0002-1839-5827, retrieved on 4 October 2026; these ten papers account for 52% of all recorded citations. Aggregate Google Scholar and Scopus indicators are reported above. References: Scopus/SciVal author profile, Scopus Author ID 57220348239; Google Scholar profile.

Part IX – Full List of Publications

Note: All papers are available on my personal website: `https://fabiofurini.github.io/publications/`

Figure 1 shows the journals hosting the 50 refereed journal papers, and Figure 2 reports the yearly output by SJR quartile of the publishing journal. Overall, 36 papers are in Q1 journals and 12 in Q2 journals; the newly accepted paper in INFORMS Journal on Optimization is shown separately pending quartile classification.

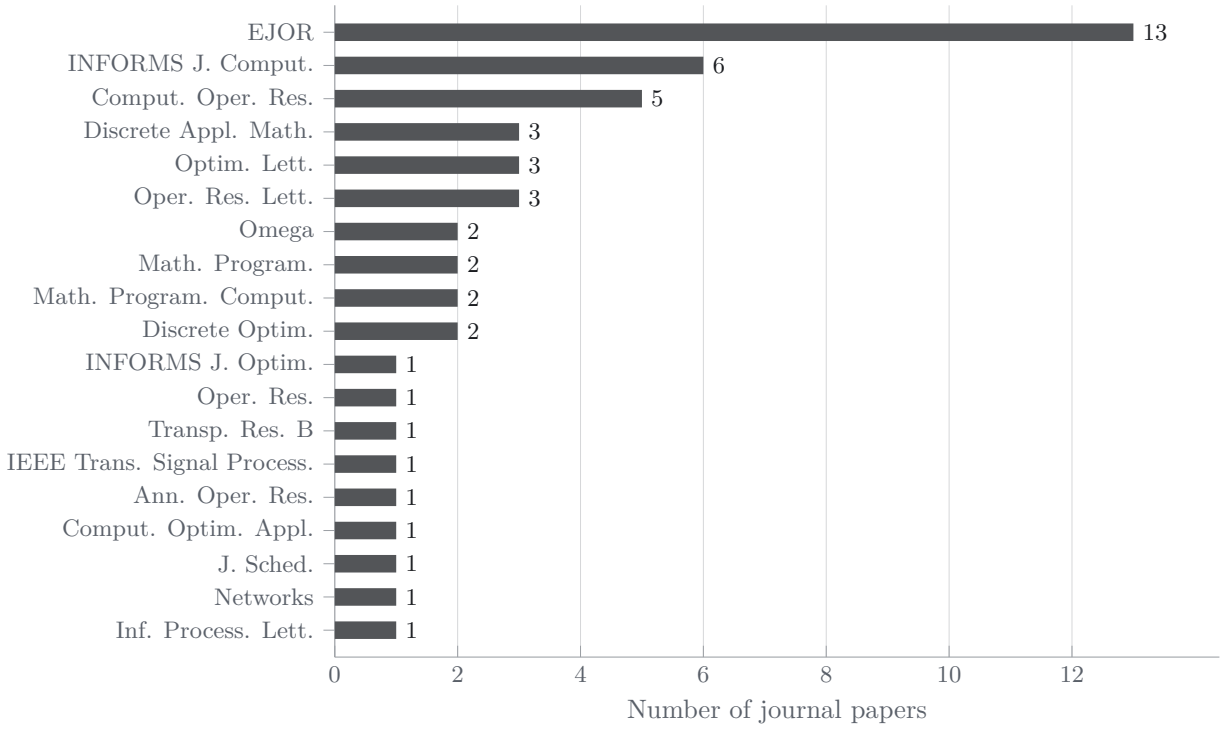


Figure 1: Journals by number of papers: the 19 journals hosting the 50 refereed journal papers, including accepted papers.

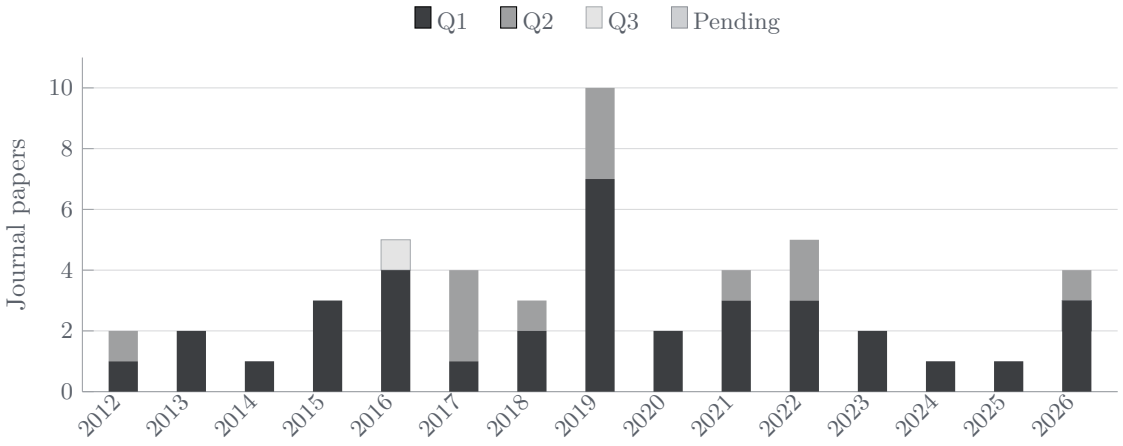


Figure 2: Journal papers per year, by SJR quartile of the publishing journal (Scimago, 2025 edition, best subject category). Of the 50 refereed journal papers, 36 are in Q1 journals and 12 in Q2 journals; one newly accepted paper awaits quartile classification.

[50] **The Bin Packing Problem with Setups: Formulations, Structural Properties and Computational Insights**, R. Baldacci, F. Ciccarelli, S. Coniglio, V. Dose, F. Furini, *2026, accepted for publication.*
European Journal of Operational Research, Elsevier.
Preprint: `https://arxiv.org/abs/2509.10075`

[49] **Branch-and-Price Strikes Back for the k-Vertex Cut Problem**, F. Ciccarelli, F. Furini, C. Hojny, M.E. Lübbecke, *2026, accepted for publication.*
INFORMS Journal on Optimization, INFORMS.

[48] **Strength of the Upper Bounds for the Edge-Weighted Maximum Clique Problem**, F. Ciccarelli, V. Dose, F. Furini, M. Monaci, *2026, accepted for publication.*
Discrete Applied Mathematics, Elsevier.

[47] **An Exact Approach for the Train Single-Routing Selection Problem**, A.L. Croella, F. Furini, I. Ljubic, B. Pascariu, P. Pellegrini, P. San Segundo, *2026, accepted for publication.*
European Journal of Operational Research, Elsevier.

[46] **Exact Algorithms for Two-Dimensional Knapsack Problems: A Unified Framework with New Benchmark Results**, S. Wang, R. Baldacci, F. Furini, L. Wei, Q. Liu, *2026, published online (Articles in Advance).*
INFORMS Journal on Computing, INFORMS. · doi.org/10.1287/ijoc.2025.1423

[45] **Integer linear programming formulations for the maximum flow blocker problem**, I. Bentoumi, F. Furini, A.R. Mahjoub, S. Martin, *2025 (17 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2025.02.013

[44] **A Numerically Exact Algorithm for the Bin-Packing Problem**, R. Baldacci, S. Coniglio, F. Furini, *2024 (21 pages).*
INFORMS Journal on Computing, INFORMS. · doi.org/10.1287/ijoc.2022.0257

[43] **CliSAT: a new exact algorithm for hard maximum clique problems**, P. San Segundo, F. Furini, D. Alvarez, P. Pardalos, *2023 (18 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2022.10.028

[42] **A Combinatorial Flow-based Formulation for Temporal Bin Packing Problems**, J. Martinovic, N. Strasdat, J. Valério de Carvalho, F. Furini, *2023 (21 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2022.10.012

[41] **Submodular maximization of concave utility functions composed with a set-union operator with applications to maximal covering location problems**, S. Coniglio, F. Furini, I. Ljubic, *2022 (48 pages).*
Mathematical Programming, Springer. · doi.org/10.1007/s10107-022-01884-7

[40] **Variable and constraint reduction techniques for the temporal bin packing problem with fire-ups**, J. Martinovic, N. Strasdat, J. Valerio de Carvalho, F. Furini, *2022 (26 pages).*
Optimization Letters, Springer. · doi.org/10.1007/s11590-021-01825-x

[39] **A new branch-and-filter exact algorithm for binary constraint satisfaction problems**, P. San Segundo, F. Furini, R. León, *2022 (20 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2021.09.014

[38] **On the exact separation of cover inequalities of maximum-depth**, D. Catanzaro, S. Coniglio, F. Furini, *2022 (21 pages).*
Optimization Letters, Springer. · doi.org/10.1007/s11590-021-01741-0

[37] **A branch-and-cut algorithm for the Edge Interdiction Clique Problem**, F. Furini, I. Ljubic, P. San Segundo, Y. Zhao, *2021 (15 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2021.01.030

[36] **Casting Light on the Hidden Bilevel Combinatorial Structure of the Capacitated Vertex Separator Problem**, F. Furini, I. Ljubić, E. Malaguti, P. Paronuzzi, *2022 (22 pages).*
Operations Research, INFORMS. · doi.org/10.1287/opre.2021.2110

[35] **A Branch-and-Price Algorithm for the Minimum Sum Coloring Problem**, D. Delle Donne, F. Furini, E. Malaguti, R. Wolfler Calvo, *2021 (18 pages).*
Discrete Applied Mathematics, Elsevier. · doi.org/10.1016/j.dam.2020.08.031

[34] **A new combinatorial branch-and-bound algorithm for the Knapsack Problem with Conflicts**, S. Coniglio, F. Furini, P. San Segundo, *2021 (21 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2020.07.023

[33] **A Branch-and-Price Framework for Decomposing Graphs into Relaxed Cliques**, T. Gschwind, S. Irnich, F. Furini, R. Wolfler Calvo, *2021 (21 pages).*
INFORMS Journal on Computing, INFORMS. · doi.org/10.1287/ijoc.2020.0984

[32] **A branch-and-price algorithm for the temporal bin packing problem**, M. Dell'Amico, F. Furini, M. Iori, *2020 (16 pages).*

Computer & Operations Research, Elsevier. · doi.org/10.1016/j.cor.2019.104825

[31] **A lexicographic pricer for the fractional bin packing problem**, S. Coniglio, F. D'Andreagiovanni, F. Furini, *2019 (7 pages).*
Operations Research Letters, Springer. · doi.org/10.1016/j.orl.2019.10.011

[30] **A note on selective line-graphs and partition colorings**, D. Cornaz, F. Furini, E. Malaguti, A. Santini, *2019 (4 pages).*
Operations Research Letters, Springer. · doi.org/10.1016/j.orl.2019.08.005

[29] **On Integer and Bilevel Formulations for the k-Vertex Cut Problem**, F. Furini, I. Ljubic, E. Malaguti, P. Paronuzzi, *2020 (32 pages).*
Mathematical Programming Computation, Springer. · doi.org/10.1007/s12532-019-00167-1

[28] **A new branch-and-bound algorithm for the Maximum Weighted Clique Problem**, P. San Segundo, F. Furini, J. Artieda, *2019 (15 pages).*
Computer & Operations Research, Elsevier. · doi.org/10.1016/j.cor.2019.05.017

[27] **A new branch-and-bound algorithm for the maximum edge-weighted clique problem**, P. San Segundo, S. Coniglio, F. Furini, I. Ljubic, *2019 (15 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2019.03.047

[26] **The Maximum Clique Interdiction Problem**, F. Furini, I. Ljubic, S. Martin, P. San Segundo, *2019 (16 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2019.02.028

[25] **Benders Decomposition for Very Large Scale Partial Set Covering and Maximal Covering Location Problems**, J.F. Cordeau, F. Furini, I. Ljubic, *2019 (15 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2018.12.021

[24] **Theoretical and computational study of several linearisation techniques for Binary Quadratic Problems**, F. Furini, E. Traversi, *2019 (25 pages).*
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[23] **QPLIB: A Library of Quadratic Programming Instances**, F. Furini, E. Traversi, P. Belotti, A. Frangioni, A. Gleixner, N. Gould, L. Liberti, A. Lodi, R. Misener, H. Mittelmann, N. Sahinidis, S. Vigerske and A. Wiegele, *2019 (29 pages).*
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[22] **The Vertex k-cut Problem**, D. Cornaz, F. Furini, M. Lacroix, E. Malaguti, A. R. Mahjoub, S. Martin, *2019 (21 pages).*
Discrete Optimization, Elsevier. · doi.org/10.1016/j.disopt.2018.07.003

[21] **An Exact Algorithm for the Partition Coloring Problem**, F. Furini, E. Malaguti and A. Santini, *2018 (12 pages).*
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[20] **On the Product Knapsack Problem**, C. D'Ambrosio, F. Furini, M. Monaci, E. Traversi, *2018 (22 pages).*
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[19] **Tighter MIP models for Barge Container Ship Routing**, L. Alfandari, T. Davidovic, F. Furini, I. Ljubic, V. Maras and S. Martin, *2019 (17 pages).*
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[18] **Exact Approaches for the Knapsack Problem with Setups**, F. Furini, M. Monaci, E. Traversi, *2018 (12 pages).*
Computer & Operations Research, Elsevier. · doi.org/10.1016/j.cor.2017.09.019

[17] **Improving the Approximated Projected Perspective Reformulation by Dual Information**, A. Frangioni, F. Furini, C. Gentile, *2017 (5 pages).*
Operations Research Letters, Elsevier. · doi.org/10.1016/j.orl.2017.08.001

[16] **An effective dynamic programming algorithm for the minimum-cost maximal knapsack packing problem**, F. Furini, I. Ljubić, M. Sinnl, *2017 (10 pages).*
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[15] **An Improved DSATUR-Based Branch-and-Bound Algorithm for the Vertex Coloring Problem**, F. Furini, V. Gabrel, I. C. Ternier, *2017 (17 pages).*
Networks, Wiley. · doi.org/10.1002/net.21716

[14] **Solving Vertex Coloring Problems as Maximum Weight Stable Set Problems**, D. Cornaz, F. Furini, E. Malaguti, *2017 (12 pages).*
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[13] **The Time Dependent Traveling Salesman Planning Problem in Controlled Airspace**, F. Furini, C.A. Persiani, P. Toth, *2016 (17 pages).*
Transportation Research Part B, Elsevier. · doi.org/10.1016/j.trb.2016.04.009

[12] **Modeling Two-Dimensional Guillotine Cutting Problems via Integer Programming**, F. Furini, E. Malaguti, D. Thomopulos, *2016 (15 pages).*
INFORMS Journal on Computing, INFORMS. · doi.org/10.1287/ijoc.2016.0710

[11] **Approximated perspective relaxations: a project and lift approach**, A. Frangioni, F. Furini, C. Gentile, *2016 (30 pages).*
Computational Optimization and Applications, Springer. · doi.org/10.1007/s10589-015-9787-8

[10] **Solving the Temporal Knapsack Problem via Recursive Dantzig–Wolfe Reformulation**, A. Caprara, F. Furini, E. Malaguti, E. Traversi, *2016 (7 pages).*
Information Processing Letters, Elsevier. · doi.org/10.1016/j.ipl.2016.01.008

[9] **Approaches to a real-world train timetabling problem in a railway node**, V. Cacchiani, F. Furini and M. P. Kidd, *2016 (14 pages).*
Omega, Elsevier. · doi.org/10.1016/j.omega.2015.04.006

[8] **Heuristic and exact algorithms for the interval min-max regret knapsack problem**, F. Furini, M. Iori, S. Martello, M. Yagiura, *2015 (14 pages).*
INFORMS Journal on Computing, INFORMS. · doi.org/10.1287/ijoc.2014.0632

[7] **Improved rolling horizon approaches to the aircraft sequencing problem**, F. Furini, M. P. Kidd, C. Persiani, P. Toth, *2015 (13 pages).*
Journal of Scheduling, Springer. · doi.org/10.1007/s10951-014-0415-8

[6] **Automatic Dantzig-Wolfe Reformulation of Mixed Integer Programs**, M. Bergner, A. Caprara, A. Ceselli, F. Furini, M. E. Lübbecke, E. Malaguti, E. Traversi, *2015 (34 pages).*
Mathematical Programming, Springer. · doi.org/10.1007/s10107-014-0761-5

[5] **Generation of Antipodal Random Vectors With Prescribed Non-Stationary 2-nd Order Statistics**, A. Caprara, F. Furini, A. Lodi, M. Mangia, R. Rovatti and G. Setti, *2014 (10 pages).*
IEEE Transactions on Signal Processing, IEEE. · doi.org/10.1109/TSP.2014.2302737

[4] **Models for the Two-Dimensional Two-Stage Cutting Stock Problem with Multiple Stock Size**, F. Furini, E. Malaguti, *2013 (10 pages).*
Computer & Operations Research, Elsevier. · doi.org/10.1016/j.cor.2013.02.026

[3] **Uncommon Dantzig-Wolfe Reformulation for the Temporal Knapsack Problem**, A. Caprara, F. Furini, E. Malaguti, *2013 (12 pages).*
INFORMS Journal on Computing, INFORMS. · doi.org/10.1287/ijoc.1120.0521

[2] **A Column Generation Heuristic for the Two-Dimensional Two-Staged Guillotine Cutting Stock Problem with Multiple Stock Size**, F. Furini, E. Malaguti, R. Medina Durán, A. Persiani, P. Toth, *2012 (10 pages).*
European Journal of Operational Research, Elsevier. · doi.org/10.1016/j.ejor.2011.10.018

[1] **Exact Weighted Vertex Coloring via Branch-and-Price**, F. Furini, E. Malaguti, *2012 (8 pages).*
Discrete Optimization, Elsevier. · doi.org/10.1016/j.disopt.2012.03.002

Published papers in conference proceedings

Conference (Acronym)	Papers
CPAIOR	1
CoDIT	2
INOC	3
IPCO	1
ISCO	3
MIC	1
SEA	1
TOTAL	**12**

Note: All papers are available on my personal website: https://fabiofurini.github.io/publications/

[12] **A Branch-and-Benders-Cut Approach to Solve the Maximum Flow Blocker Problem**, I. Bentoumi, F. Furini, A.R. Mahjoub, S. Martin, *9th International Conference on Control, Decision and Information Technologies (CoDIT 2023), Rome, Italy, July 2023 (4 pages).*
Publisher: IEEE Proceedings.
doi.org/10.1109/CoDIT58514.2023.10284073

[11] **ILP Models and Column Generation for the Minimum Sum Coloring Problem**, F. Furini, E. Malaguti, S. Martin and I.-C. Ternier, *8th International Network Optimization Conference (INOC 2017), Lisbon, Portugal, February 2017 (10 pages).*
Publisher: Electronic Notes in Discrete Mathematics. Elsevier.
doi.org/10.1016/j.endm.2018.01.023

[10] **MIP Formulations for a Rich Real-world Lot-sizing Problem with Setup Carryover**, F. Focacci, F. Furini, V. Gabrel, D. Godard and X. Shen, *5th International Symposium on Combinatorial Optimization (ISCO 2016), Vietri sul Mare, Italy, May 2016 (12 pages).*
Publisher: Lecture Notes Computer Science, Springer.
doi.org/10.1007/978-3-319-45587-7_11

[9] **Matheuristics for the Temporal Bin Packing Problem**, F. Furini and X. Shen, *11th Metaheuristics International Conference (MIC 2015), Agadir, Morocco, June 2015 (13 pages).*
Publisher: Springer book. ISBN: 978-3-319-58252-8.
doi.org/10.1007/978-3-319-58253-5_19

[8] **Lower Bounding Techniques for DSATUR-based Branch and Bound**, F. Furini, V. Gabrel, I. C. Ternier, *7th International Network Optimization Conference (INOC 2015), Warsaw, Poland, June 2015 (8 pages).*
Publisher: Electronic Notes in Discrete Mathematics. Elsevier.
doi.org/10.1016/j.endm.2016.03.020

[7] **ILP and CP Formulations for the Lazy Bureaucrat Problem**, I. Ljubic, F. Furini, M. Sinnl, *12th International Conference on Integration of Artificial Intelligence and Operations Research Techniques in Constraint Programming (CPAIOR 2015), Barcelona, Spain, May 2015 (15 pages).*
Publisher: Lecture Notes Computer Science, Springer.
doi.org/10.1007/978-3-319-18008-3_18

[6] **Mathematical Formulations for the Balanced Vertex k -Separator Problem**, D. Cornaz, F. Furini, M. Lacroix, E. Malaguti, A. R. Mahjoub, S. Martin, *International Conference on Control, Decision and Information Technologies (CoDIT 2014), Metz, France, November 2014 (8 pages).*

Publisher: IEEE Conference.

doi.org/10.1109/CoDIT.2014.6996889

[5] **State space reduced dynamic programming for the aircraft sequencing problem with constrained position shifting**, F. Furini, M. P. Kidd, A. Persiani, P. Toth, *4th International Symposium on Combinatorial Optimization (ISCO 2014), Lisbon, Portugal, March 2014 (12 pages).*

Publisher: Lecture Notes Computer Science, Springer.

doi.org/10.1007/978-3-319-09174-7_23

[4] **A fast heuristic approach for train timetabling in a railway node**, F. Furini, M. P. Kidd, *6th International Network Optimization Conference (INOC 2013), Tenerife, Spain, May 2013 (8 pages).*

Publisher: Electronic Notes in Discrete Mathematics. Elsevier.

doi.org/10.1016/j.endm.2013.05.094

[3] **Hybrid SDP Bounding Procedure**, F. Furini, E. Traversi, *12th International Symposium on Experimental Algorithms (SEA 2013), Rome, Italy, June 2013 (12 pages).*

Publisher: Lecture Notes Computer Science, Springer.

doi.org/10.1007/978-3-642-38527-8_23

[2] **Aircraft Sequencing Problems via a Rolling Horizon Algorithm**, F. Furini, C. Persiani, P. Toth, *3rd International Symposium on Combinatorial Optimization (ISCO 2012), Athens, Greece, April 2012 (12 pages).*

Publisher: Lecture Notes Computer Science, Springer.

doi.org/10.1007/978-3-642-32147-4_25

[1] **Partial convexification of general MIPs by Dantzig-Wolfe reformulation**, M. Bergner, A. Caprara, F. Furini, M.E. Lübbecke, E. Malaguti, E. Traversi, *15th International Conference on Integer Programming and Combinatorial Optimization (IPCO 2011), New York, USA, June 2011 (12 pages).*

Publisher: Lecture Notes Computer Science, Springer.

doi.org/10.1007/978-3-642-20807-2_4

Selected working papers

[1] **A Highly Efficient Numerically Exact Algorithm for Two-Dimensional Bin-Packing Problems**, S. Wang, R. Baldacci, F. Furini, L. Wei, Q. Liu, *2025, under revision.*

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5515822

[2] **A first approximation algorithm for the Bin Packing Problem with Setups**, R. Baldacci, F. Ciccarelli, S. Coniglio, V. Dose, F. Furini, *2025.*

https://arxiv.org/abs/2512.24785

[3] **A Two-Phase Matheuristic for the Integrated Healthcare Timetabling Problem**, F. Ciccarelli, A. Di Biase, F. Furini, *2025.*

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5768118

[4] **Optimal Macroitem Sequences in the Precedence Constrained Knapsack Problem**, V. Dose, F. Furini, M. Locatelli, *2026.*

https://arxiv.org/abs/2606.22018

[5] **An Integer Programming Approach to Compute Lower Bounds for Ramsey Numbers Using Circulant Graphs**, S. Coniglio, F. Furini, I. Ljubić, P. San Segundo, J. Thüerauf, E. Traversi, *2026.*

https://arxiv.org/abs/2608.18769

Part X – Editorial, Reviewing and Scientific Service

2018–now **Associate Editor, Omega (The International Journal of Management Science)**, Elsevier.

2018–2020 **Guest Editor of the special issue of Discrete Applied Mathematics**, titled "Graphs and Combinatorial Optimization", Elsevier.

2017–2019 **Guest Editor of the special issue of Annals of Operations Research**, titled "Recent Advances in Decomposition Methods for Hard Optimization Problems", Elsevier.

Referee activities

2009–now **Referee activity**, I regularly serve as a referee for leading international journals, including INFORMS Journal on Computing; European Journal of Operational Research; Journal of Scheduling; Computers & Operations Research; Computers & Industrial Engineering; Journal of Optimization Theory and Applications; Journal of the Operational Research Society; RAIRO–Operations Research; Operations Research; Computational Optimization and Applications; Omega (The International Journal of Management Science); Discrete Optimization; Discrete Applied Mathematics; OR Spectrum; Mathematical Programming; and Transportation Science.

2013–now **Reviewer for research agencies**, I regularly review proposals for international research agencies, including the Agence Nationale de la Recherche (ANR), the Chilean National Science and Technology Commission, and NSERC (Civil, Industrial and Systems Engineering).

Program committees and invited sessions

2026 **MIP Program Committee Chair and local organizer**, Mixed Integer Programming European Workshop (EUROMIP), *Sapienza University of Rome, Italy, October 19–21.* `https://www.mixedinteger.org/EUROMIP/2026/`

2020 **Local Organization Committee**, CTW 2020, online.

2018 **Local Organization Committee**, CTW 2018 in Paris (CNAM).

2018 **Local Organization Committee**, SPOC18 in Paris (Université Paris Dauphine-PSL).

2018 **Organizer of invited sessions on "Combinatorial Optimization"**, Joint EURO/ALIO International Conference 2018 on Applied Combinatorial Optimization, Bologna, June 25-27, 2018.

2018 **Organizer of invited sessions on "Operations Research and Applications"**, ISMP International Conference, Bordeaux, June 1-6, 2018.

2015 **Organizer of invited sessions on "Graph Theory"**, OR 2015 International Conference, Vienna, September 10-15, 2015.

Part XI – Collaborations, Visiting Positions and Presentations

Italy **University of Bologna**, Professors Paolo Toth, Enrico Malaguti, Andrea Lodi, Silvano Martello, Valentina Cacchiani and Michele Monaci.

University of Bergamo, Professor Stefano Coniglio.

University of Pisa, Professor Antonio Frangioni.

CNR IASI, Professor Claudio Gentile.

France **Université Paris Dauphine-PSL**, Professor A. Ridha Mahjoub.

Sorbonne Paris Nord University, Professor Roberto Wolfler Calvo.

École Polytechnique, Professor Leo Liberti and Claudia D'Ambrosio.

ESSEC Business School, Professors Emiliano Traversi, Laurent Alfandari and Diego Delle Donne.

Gustave Eiffel University, Professor Paola Pellegrini.

Austria **WU Vienna (Vienna University of Economics and Business)**, Professor Ivana Ljubić.

Spain **Universidad Politecnica de Madrid**, Professor Pablo San Segundo.

Belgium **Université Catholique de Louvain**, Professor Daniele Catanzaro.

Qatar **Hamad Bin Khalifa University**, Professor Roberto Baldacci.

China **Guangdong University of Technology**, Professors Sunkanghong Wang, Lijun Wei, Qiang Liu.

Germany **RWTH Aachen University**, Professor Marco Lübbecke.

University of Mainz, Professor Stefan Irnich.

Japan **Nagoya University**, Professor Mutsunori Yagiura.

Visiting positions

2019 **University of Parma**, Visiting Scientist: Dipartimento di Ingegneria e Architettura, *four months of research with Professor M. Locatelli.*

2018 **University of Southampton**, Visiting Scientist: Faculty of Mathematics, *one month of research with Professor S. Coniglio.*

2018 **Accueil en délégation institutionnelle CNRS**, semester of research at Sorbonne University (formerly Pierre and Marie Curie University, Paris 6).

2014 **University of Vienna**, Visiting Scientist: Faculty of Business, Economics, and Statistics, *one month of research with Professor I. Ljubic.*

2012 **University of Colorado**, Visiting Scientist: Leeds School of Business, *two months of research with Professors M. Laguna and F. Glover.*

2011 **RWTH Aachen University**, Visiting Scientist: Operations Research Group, *three months of research with Professor M. Lübbecke.*

2010 **Imperial College London**, Visiting Scientist: Imperial College Business School, *six months of research with Professor E. Hadjiconstantinou.*

2007 **University College Dublin**, Visiting Scientist: Quinn School of Business, *six months of research with Professor C. Brugha.*

Invited seminar presentations

2025 **Advanced mathematical formulations for Graph Disconnection Problems**, RWTH Aachen University, Aachen, Germany.

2025 **Hidden Bilevel Structures in Graph Disconnection Problems: Models and Exact Algorithms**,
Dipartimento di Scienze Economiche, University of Bergamo, Italy.

2024 **Submodular maximization of concave utility functions composed with a set-union operator with applications to maximal covering location problems**,
Department of Mathematics, University of Trier, Germany.

2021 **Decomposition techniques for hard combinatorial optimization problems**, Department of Computer, Control and Management Engineering, Sapienza University of Rome.

2019 **Interdiction problems on graphs**, Dipartimento di Ingegneria e Architettura, University of Parma.

2018 **Partial Set Covering via Benders Decomposition**, Centre for logistics and supply chain management, University of Luxembourg.

2018 **Most vital nodes with respect to the Clique Number**, IASI CNR, Rome.

2018 **Clique Interdiction Games**, University of Southampton, UK.

2018 **Reformulations and Decompositions of Mixed Integer Linear Programs**, ESSEC Business School, Paris.

2016 **BQP formulations and relaxations**, CNAM, Paris.

2015 **Dantzig-Wolfe Reformulation for Generic MIPs**, Université de la Lorraine, Metz.

2014 **Selected Topics in Quadratic Programming**, University of Vienna, Vienna.

2013 **Hybrid LP/SPD Bounding Procedure**, École Polytechnique, Paris.

2013 **Extended Linear Formulation for Binary Quadratic Problems**, Sorbonne Paris Nord University (formerly Université Paris 13), Paris.

2012 **Min-max and min-max regret optimization**, TU Dortmund.

2010 **Column Generation for the Temporal Knapsack Problem**, Southampton University.

2010 **Decomposition of Integer Linear Programming Problems**, RWTH Aachen University.

2010 **Pseudo Compact Reformulation for the Resource Allocation Problem**, University of Liège, HEC-Management School.

Selected conference presentations

2026 **Hidden Bilevel Structures in Graph Disconnection Problems**, *24th Conference of the International Federation of Operational Research Societies (IFORS 2026), Vienna, Austria, July.*

2026 **Plenary speaker**, *DOC2026 and YPOC2026 workshops, Cagliari, Italy, May.* `https://sites.google.com/view/doc2026/home-page`

2025 **Hidden Bilevel Structures in Graph Disconnection Problems: Stackelberg Game Models and Exact Algorithms (invited speaker)**, *Mixed Integer Programming European Workshop (EUROMIP), Clermont-Ferrand, France, July 1–3.*

2024 **Location covering problems**, *33rd European Conference on Operational Research (EURO), Copenhagen, Denmark, June–July.*

2024 **Maximal covering location problems**, *25th International Symposium on Mathematical Programming (ISMP), Montréal, Canada, July.*

2024 **A new model for the train routing selection problem**, *International Conference on Optimization and Decision Science (ODS), Badesi, Italy, September.*

2023 **The sharpest bin-packing column (invited speaker)**, *Column Generation (CG), Montréal, Canada, May.*

2022 **Submodular maximization of concave utility functions composed with a set-union operator with applications to maximal covering location problems**, *19th Workshop on Advances in Continuous Optimization (EUROPT), Caparica, Portugal, July.*

2019 **Large scale Partial Set Covering Location Problems**, *30th European Conference on Operational Research (EURO), Dublin, Ireland, June.*

2019 **Clique Interdiction Games on Graphs**, *23rd International Symposium on Mathematical Programming (ISMP), Bordeaux, France, July.*

2018 **The Maximum Clique Interdiction Games**, *The Aussois Combinatorial Optimization Workshop, Aussois, France, January.*

2017 **Exact Algorithms for the Knapsack Problem with setups**, *Optimization and Decision Science Conference (ODS), Sorrento, Italy, September.*

2016 **QPLIB: A Library of Quadratic Programming Instances**, *5th International Conference on Continuous Optimization (ICCOPT), Tokyo, Japan, August.*

2015 **Extended Formulations for the Graph Partitioning Problems**, *International Conference on Operations Research (OR), Vienna, Austria, September.*

2015 **Approximated perspective relaxations: a project and lift approach**, *22nd International Symposium on Mathematical Programming (ISMP), Pittsburgh, USA, July.*

2010 **Column Generation Approach for AGV dispatching problem in port terminal**, *24th European Conference on Operational Research (EURO), Lisbon, Portugal, July.*

2009 **Practical solution of the Resource Allocation Problem**, *AIRO (Italian Operations Research Society Conference), Salerno, Italy, September.*

Part XII – Supervision and Institutional Activities

2023–now **PhD supervision**, supervisor of PhD students in the ABRO doctoral program, Sapienza University of Rome.

[1] **Fabio Ciccarelli**, PhD thesis: "Exact algorithms for packing problems". Current position: third year PhD.

2024–now **PhD Visiting**, supervisor of PhD students visiting from different institutions.

[1] **Mingming Zheng**, PhD thesis: "Exact algorithms for graph coloring problems". Visiting PhD student at Sapienza (2024–2026, China Scholarship Council grant). Current position: PhD student in Management Science and Engineering, Huazhong University of Science and Technology (HUST), Wuhan, China.

[2] **Yuntian Zhang**, PhD thesis: "Location covering problems". Current position: third year PhD.

2021–now **Post-doc supervision**, supervisor of postdoctoral researchers, Sapienza University of Rome.

[1] **Daniele Patria**, Postdoctoral research activity: "Packing problems with submodular objective functions". Current position: Consultant at Siemens.

2021–now **Thesis supervision**, supervisor of numerous Bachelor's and Master's theses at the Department of Computer, Control and Management Engineering, Sapienza University of Rome.

2014–2024 **PhD co-supervision**, at the doctoral school of Université Paris Dauphine-PSL.

[1] **Ian-Christopher Ternier**, Thesis topics: "Résolution exacte du Problème de Coloration de Graphe et ses variantes". Current position: Professor at EPITA.

[2] **Xueying Shen**, Industrial PhD carried out through a French doctoral contract jointly involving university, company and doctoral candidate, with *DecisionBrain*. Thesis topic: "Problèmes complexes de dimensionnement de lots de production avec machines parallèles et report de configuration". Current position: DecisionBrain.

[3] **Isma Bentoumi**, Industrial PhD carried out through a French doctoral contract jointly involving university, company and doctoral candidate, with *Huawei*. Thesis topic: "Étude du problème de bloqueur appliqué aux multiflots : analyse polyédrale et algorithmes". Current position: GIM Dassault.

2013–2019 **Thesis supervision**, supervisor of numerous Master's theses at LAMSADE, Université Paris Dauphine-PSL.

2011–2012 **Thesis supervision**, supervisor of Bachelor's and Master's theses at the Department of Electrical, Electronic and Information Engineering, University of Bologna.

Administrative activities

2022–now **Member of the Academic Board, PhD Program in Automatic Control, Bioengineering and Operations Research (ABRO)**, Department of Computer, Control and Management Engineering, Sapienza University of Rome.

2023–now **Responsible for internship validation for Management Engineering students**, Bachelor's and Master's degree programmes in Management Engineering, Sapienza University of Rome.

2017–2019 **Steering committee for the organization of research seminars**, Laboratoire d'Analyse et Modélisation de Systèmes pour l'Aide à la Décision (LAMSADE), Université Paris Dauphine-PSL, UMR-CNRS 7243.

2015–2017 **Head of Master MIAGE-IF (Informatique pour la Finance)**, Université Paris Dauphine-PSL, France. Responsible for the academic and administrative management of the professional master's program in Computer Science for Finance (MIAGE-IF), including curriculum design, student recruitment, relations with industry partners, and quality assurance. The program combines computer science, mathematics, and finance to train specialists in financial information systems and quantitative finance. Annual enrollment: 30 students. The program was ranked as one of the best master's programs in its category in France.

2014-2016 **Member of the CCR of LAMSADE**, at Université Paris Dauphine-PSL, France.

Rome, November 7, 2026.